

qml by klnsss for msrit Beng 2826 day1

for Msrit by sarma

2nd August 2026

Refresher on quantum computing w.r.to QML

so, to start with one needs:

- benefits and concerns / Limitations of QC
- Mathematical foundation for Qc,QML, right from 'complex numbers'
 - 'quantum parallelism'
 - 'quantum speedup'
 - 'quantum advantage'
- Notion of state vector
- 1 qubit states (including 'gates',)

Usefulness of quantum computing / necessity(relevance of) of QC

- called 'quantum advantage'
- **quantum advantage**— improvement as compared to classical computing for the same task.
- 'quantum advantage' enables one to compare / assess:
- QC uses
- any specific limitations of classical computing
- particular Hybrid approaches, i.e 'quantum inspired' 'quantum classical' (hybrid) approaches / applications.

comparison in terms of speed up, power, input size, distributed loads, superior hardware config.

- only in limited cases computing resource ($\uparrow$) $\implies$ performance increase.
- in certain other cases, as i/ p size increases linearly $\implies$ time , memory increase exponentially
- where as
- quantum solutions / computing requires only polynomial increase in time and memory.
- $\therefore$ quantum computing / algorithms $\implies$ scalable solutions for complex tasks / problems.
- in specific cases:
- Quantum computers ONLY perform and classical computers are of no use

which one to use and when / deciding factors

in situations / tasks like

- Random sampling from sampling distributions.
- simulations of 'physical and chemical' processes

2 primary Factors contributing to 'quantum advantage'

- quantum parallelism
- quantum measurement (implementation mechanism for conditional transformations)

here comes the Necessity to also mention

- Quantum superposition
- Entanglement

Quantum interference is relevant in many quantum algorithms. quantum measurements are restricted to specific discrete values (Eigen values of the operator)

Quantum interference def'n / notion

"Phenomenon in which particles / qubits / photons exist in superposition states and interfere with themselves upon measurement"

a useful but essential digression / deviation / why complex numbers

- Quantum states— are described by "Normalized complex vector in a vector space"

Back to Quantum superposition

- Quantum superposition
- $\Downarrow$
- quantum system can exist in multiple states
- $\Downarrow$
- computationally, quantum state(system) contains all probabilities of all possible outcomes
- $\Downarrow$
- probabilities change instantly + simultaneously (in effect quantum parallelism) $\Leftarrow$ computational consequence of "quantum superposition"

Implications of quantum Entanglement ↓↓

- Quantum bits (qubits) are connected in such a way that state can't be written as a 'product state (tensor product)'
- ↓↓
- states cant be independently described, of each other
- ↓↓
- independent qubit measurements are not independent of each other
- ↓↓
- practical mechanism of "conditional transformation"
- *therefore* for non-trivial transformations, "quantum entanglement is a must " for most of quantum computing

- as a simple use case— say —"pair swapping" in a list (to SWAP even and odd values)
- ↓↓
- kind of linearity even in case of "no.of operations", effectively like SIMD—"single-instruction multiple data"
- but, in case of Quantum computation
- Quantum implementation requires only one instruction to swap the all required pairs.
- ↓↓
- Unlimited elementary parallel operations= "quantum parallelism"— consequence of 'quantum superposition'

- quantum advantage
- $\Updownarrow$
- quantum parallelism
- but, in certain specific problems
- same instruction for all pairs is unwanted
- recombination of value (probability) pairs occurs simultaneous
- $\Downarrow$
- Quantum advantage

consequences of Randomness of quantum measurement

- Measurement in quantum system
- ↓↓
- outcome is non deterministic
- ↓↓
- one of the defined set of outcomes
- obeying / following certain probability distribution
- ↓↓
- context by context (depending on problem)— repeated measurements yield useful information from the distribution of outcomes

inevitability of measurement !

- quantum state 'values' define "probability of selecting each outcome"
- so, values of the "quantum state " can't be inspected directly
- ↓↓
- quantum measurement — is like just sampling from probability distribution
- so, to know the values / infer , set of measurements are required.
- more often than not (in majority contexts)
"Measurements" are beneficial
- ↓↓
- "so much that" measurements enable computations, otherwise not possible with — classical computation.
- one such simple example is— "random number generation"

Deciding factors for Quantum approach

- concept of parallelism
- interpretation of "Measurement"

- single qubit computation
- Multiple qubit computation, includes
- 2 qubit computation
- 3 qubit computation
- qubits are usually Encoded by photons / electrons / few more(further not relevant for us)
- computation / programming perspective—qubit — is a probabilistic bit

so, in , computation with qubit / probabilistic bit

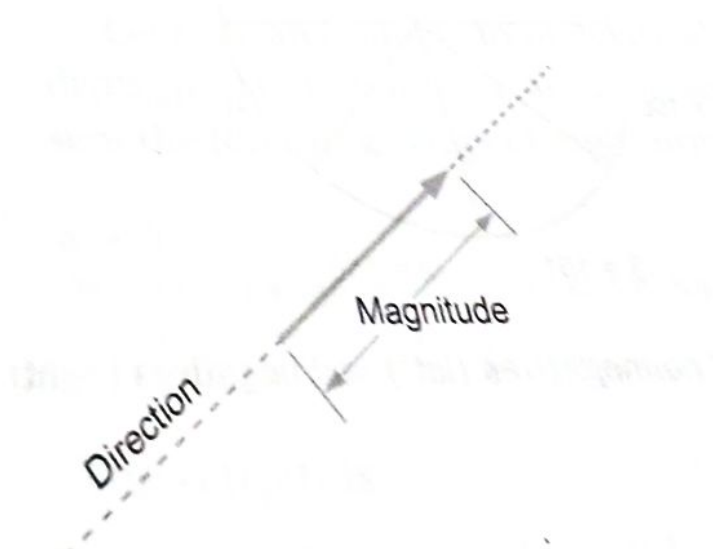
- in order to get "one of the out comes" by "Measurement"
"probability of out comes is changed"
- probability is the squared length of planar vector
- name of such probability in quantum computing is
- Amplitude (s)
- Complex numbers mathematically
- $\Downarrow$
- in QC, complex numbers $\implies$ Amplitudes
- direction of Amplitudes is important / essential in knowing how "quantum system changes"

3 methods to represent Amplitudes

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- As arrows inside the unit circle (2 dimensions)
- Algebraic representation (as complex numbers)
- polar form (Trigonometric representation)

Amplitude inside a unit circle



from a specific publisher with humble thanks

Algebraic form of Amplitude

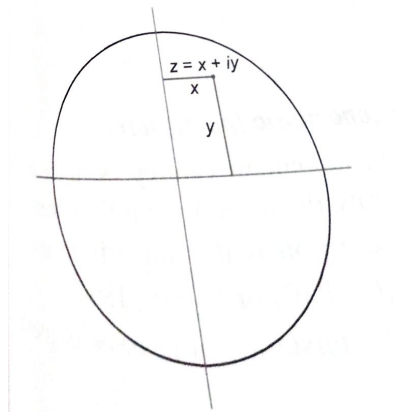


Figure: from a published book with thanks

polar form of complex number

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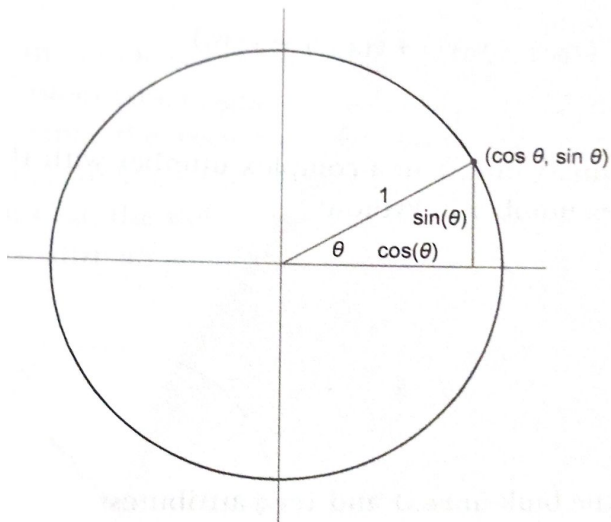


Figure: thanks to publisher /author

Further clarity on complex number rep'n (3 slides)

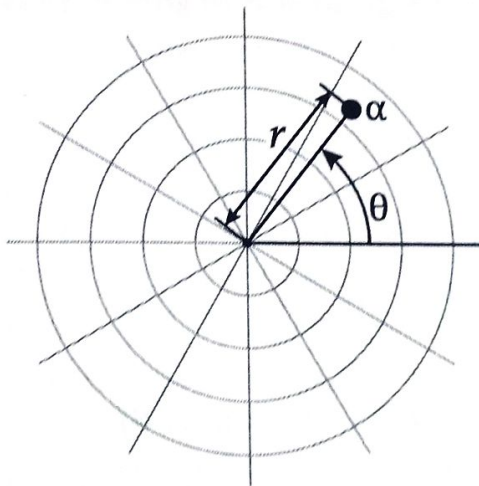


Figure: Author / publisher–acknowledgment with thanks

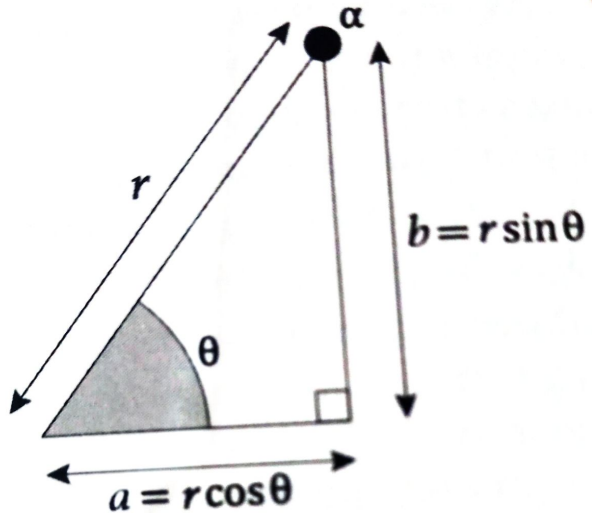


Figure: with thanks to the publisher / author

Given $a + bi$	Given (r, θ)
$r = \sqrt{a^2 + b^2} $	$a = r \cos \theta$
$\theta = \tan^{-1}(b / a)$	$b = r \sin \theta$

Figure: with thanks to the publisher / author

Argand diagram rep'n of complex number, similar to algebraic

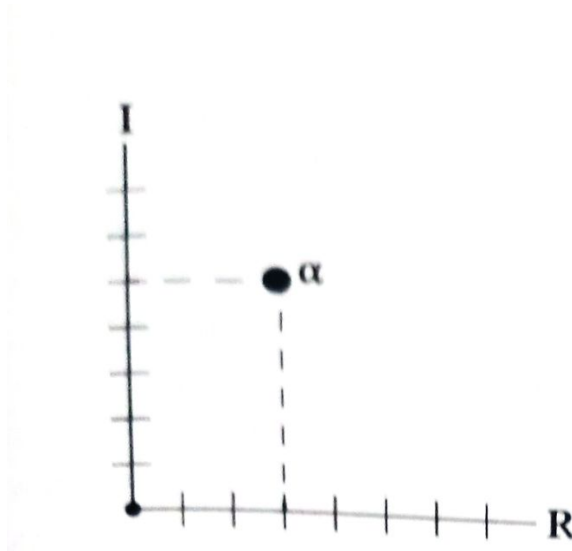


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legend / notes / clarity w.r.to above slides notation

- in the Argand diagram,
- origin— where Real (R) and Imaginary axes meet
- 'r'— magnitude of complex number (always 0 or +ve)
- ' θ '—Phase
- θ — counterclockwise from +ve side of 'x' axis
- also $r = re^{i\theta} = r \cos \theta + ir \sin \theta$
- $e^{i\pi+1} = 0$ —Euler's identity

summary of quantum computing model

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- "pairs of probabilities are changed simultaneously, computation outcomes are picked / selected acc.to these probabilities"
 - state of quantum computation consists of a probability— with direction for each outcome(possible)
 - basic computing instructions in QC $\xrightarrow{\text{chnge}}$ pairs of probabilities with direction in parallel
 - data available from QC= "Measurement outcome"
 - before measurement $\implies$ state of superposition
 - after measurement $\implies$ state of system $\xrightarrow{\text{collapse to}}$ to a state with measured probability = '1'

3 dominant approaches of quantum computing

- sampling from probability distributions
- searching for specific outcomes
- example of specific outcome search : " digital encoding"— if qubit measured is '0' or '1' outcome can be encoded applicable to the problem of interest.
portfolio optimization problem "— example of digital encoding, where a binary digit $\xrightarrow{\text{indicates}}$ whether investment is included in a portfolio.
"Grover's algorithm" also similar(search outcome)
- Estimating the probability of specific outcomes
- known as "Analog encoding"— probability of an outcome / combined probability of outcomes —eg: **shor's Algorithm**
- Analog and digital encodings can be converted but with hardware cost.

Aim is to look at :

- single qubit states
- single qubit gates
- transformation of a single qubit state
- $\Downarrow$
- single qubit circuits
- single qubit inverses
- examples of single qubit computations

single-qubit state(state vector Notation)

- single qubit state $\implies$ pair of complex numbers $\implies$ 2 possible outcomes
- $\implies$ pair(2) of amplitudes for each out come(2 possible outcomes)
- Amplitude / state vector notation—textbook example:
- 2 amplitudes be — $(-34+.43i)$, $(-.68+.49i)$,
- then comparing with $z=x+iy$, $|z|=\sqrt{x^2+y^2}$, θ from trigonometric form
- given amplitudes correspond to—out come '0', outcome '1', so comparing with Dirac notation $\implies$
- i.e a general qubit state $-\alpha = \alpha |0\rangle + \beta |1\rangle \implies |\alpha|^2 + |\beta|^2 = 1$, \$, *normalized(probabilities must add upto '1')*

Multiplication of complex numbers in polar form

considering 2 complex numbers:

$$z = r(\cos \theta + i \sin \theta)$$

$$c = r(\cos \phi + i \sin \phi)$$

product 'zc'

$$rs(\cos (\theta + \phi) + i \sin (\theta + \phi))$$

$\Rightarrow$

- Magnitudes are multiplied
- directions are added

$\implies$

$$-0.3415 + 0.4282i = \sqrt{0.3} \left(\cos \frac{5\pi}{7} + i \sin \frac{5\pi}{7} \right),$$

$$-0.6789 + 0.4918i = \sqrt{0.7} \left(\cos \frac{\pi}{5} + i \sin \frac{\pi}{5} \right)$$

see that by ccode / manually– Amplitudes must add upto 1
probability of outcome with amplitude 'z' = $|z|^2 = p$

single qubit state, Amplitudes and directions will be

- for outcome '0'
Amplitude = $\sqrt{p}(\cos \theta_0 + i \sin \theta_0) \implies$ direction = θ_0 ,
magnitude = $\sqrt{p}$, probability = p
- for outcome '1'
Amplitude = $\sqrt{1-p}(\cos \theta_1 + i \sin \theta_1) \implies$ direction = θ_1 ,
magnitude = $\sqrt{1-p}$, probability = $1-p$
- $\implies$
- general state will be:

$$|\psi\rangle = e^{i\theta_0} \left(\sqrt{p_0} |0\rangle + \sqrt{1-p_0} e^{i(\theta_1-\theta_0)} |1\rangle \right)$$

with $0 < \phi < 2\pi$

- $e^{i\theta_0}$ — global phase and can be just ignored

a caution with angles of Bloch sphere

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- qubit visualization requires only 2 angles—namely , θ latitude, ϕ longitude.
- in such case, if it is to be represented on a sphere, points (θ, ϕ) and $(\pi - \theta, \phi + \theta)$ represent the same qubit upto a factor of -1.
- to overcome , latitude is twice / double (sine angle) or $\frac{\theta}{2}$ is taken, so equations apparently differ (to take care of 3D projection onto 2D-sphere to circle)
- hence , $|\psi\rangle = \cos \frac{\theta}{2} |0\rangle + i\phi \sin \frac{\theta}{2} |1\rangle$

- latitude— distance to poles, $\rightarrow a, b$ with $\theta = 0$ pointing up for $|0\rangle \rightarrow (a=1, b=0)$
- $\theta = \pi \xrightarrow{\text{straightdown}} \text{for } |1\rangle (a=0, b=1)$
item and $\theta = \frac{\pi}{2} \rightarrow \text{equator (basic superposition), } (a=b=\frac{1}{\sqrt{2}})$

- for a single qubit, superposition states will be
- $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$; $|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$
- differ by -1 in $|1\rangle \rightarrow$ relative phase(angle)
- minus sign $\implies \pi(180^\circ)$ with $e^{i\pi} = -1$
- relative phases lead to **constructive interference**, **destructive interference**
- when entire state is considered (not just 1 term in superposition)— global phase $\implies$ no impact on quantum measurements

Gates="changing amplitudes"

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- quantum gates alter the pairs of amplitudes in quantum systems with any number of qubits
- quantum gates implement unitary transformations on qubits($UU^\dagger = I$), where U — unitary matrix, I — identity matrix
- unitarity ensures that quantum gates preserve total probability

- operators $\xrightarrow[\text{on}]{\text{act}}$ quantum states $\xrightarrow[\text{physical transformations}]{\text{Measurements}}$ linear transformations expressed in matrices.
- Hermitian operators $\xrightarrow{\text{represent}}$ observable quantities.
 $\implies \hat{O} = \hat{O}^\dagger$
- $\hat{O}^\dagger \rightarrow$ conjugate transpose of $\hat{O}$.
- Eigen values of a Hermitian operator = real $\implies$ Measurements give real values only
- $\hat{H} \rightarrow$ represents total energy of a quantum system for us
 $\rightarrow$ is hermitian,

Pauli matrices are(Hermitian +unitary)

- pauli matrices are crucial for comprehending single-qubit operations.
- pauli matrices $\rightarrow$ rotations about 'x','y','z' axes of Bloch sphere
- $\sigma_x \rightarrow$ flipping of qubit state i.e $|0\rangle \mapsto |1\rangle$
- $\sigma_y \rightarrow$ applies $\frac{\pi}{2}$ phase shift while flipping state.
- $\sigma_z \rightarrow$ applies a π phase shift to $|1\rangle \xrightarrow[\text{component of}]{\text{the state}}$
- these are widely used in Quantum gates.
- quantum gates are just Unitary transformations.i.e $UU^\dagger = I$

Notation of diagrams in quantum circuits

- circuit diagrams are constructed and read from left to right
- qubits $\xrightarrow{\text{represented}}$ with horizontal Lines.
- QUANTUM GATES $\xrightarrow[\text{represented as}]{\text{symbols}}$ on qubit lines
- single qubit gates $\rightarrow$ labeled boxes
- CNOT gate (example of multi qubit) $\xrightarrow{\text{drawn}}$ with connections between qubits
- Measurements $\xrightarrow[\text{represented with}]{M(\text{metersymbol})}$
- a line with no operator on it $\implies$ qubit remains in the state in which it is prepared

phase shift gate (single-qubit)–basic for QML

- phase shift gates are essential for **controlled phase rotations** in multi qubit circuits.
- phase shift gate adds a phase ' ϕ ' to $|1\rangle$ component of the qubit. $R_\phi = \begin{bmatrix} 1 & 0 \\ 0 & e^{i\phi} \end{bmatrix}$
- for specific value $\phi = \frac{\pi}{2}$, quantum state becomes / modified from $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ to $|\psi'\rangle = \alpha|0\rangle + \beta e^{i\frac{\pi}{2}}|1\rangle$

CNOT (2-qubit)–multi qubit gate relevance in qml/ Algorithms

- CNOT $\xrightarrow{\text{flips}}$ target qubit $|t\rangle$ if the control qubit $|c\rangle$ is in $|1\rangle$ state.

- $\text{CNOT}_{\text{matrix}} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$

- basis states are transformed by CNOT as follows

$$|00\rangle \mapsto |01\rangle$$

$$|01\rangle \mapsto |01\rangle$$

$$|10\rangle \mapsto |11\rangle$$

$$|11\rangle \mapsto |10\rangle$$

Relevance of controlled gates in Quantum Algorithms

- conditional operations are realized using controlled gate operations
- popular examples:
 - $\Downarrow$
 - Grover's search algorithm
 - QFT (quantum Fourier transform)
 - infact, also used in—Error correction, Entanglement generation.

- Dirac's notation includes amongst others,
- 'ket', 'bra' vectors
- inner products with 2 properties (assignment)
- outer product (operator construction— maps one state to another)
- Applications / relevance of Dirac notation include:
- quantum measurement
- quantum operations
- Tensor products (Multi-qubit states)

- $|\psi\rangle$, ket $\mapsto$ column vector in complex vector space
- as an example, single qubit state $|\psi\rangle = \begin{bmatrix} \alpha \\ \beta \end{bmatrix}$
- $\alpha, \beta \mapsto$ complex numbers subject to $|\alpha|^2 + |\beta|^2 = 1$
- $\langle\psi| = [\alpha^* \beta^*] \mapsto$ Hermitian conjugate (complex conjugate transpose) of $|\psi\rangle$
- $\implies$ 'bra'— row vector.

- $\langle \phi | \psi \rangle \mapsto$ inner product of quantum states $|\phi\rangle, |\psi\rangle$ with
 $|\psi\rangle = \begin{bmatrix} \alpha_1 \\ \beta_1 \end{bmatrix}, |\phi\rangle = \begin{bmatrix} \alpha_2 \\ \beta_2 \end{bmatrix}$
- Inner product is subject to 2 properties– Linearity, Hermitian symmetry.
- inner product=complex number, magnitude determines probability amplitude of one state transitioning to other
- Hermitian symmetry $\implies \langle \phi | \psi \rangle = (\langle \psi | \phi \rangle)^*$

outer product(ketbra) $\mapsto$ constructs an operator

- say, $|\psi\rangle = \begin{bmatrix} \alpha \\ \beta \end{bmatrix}$, $\langle\phi| = \begin{bmatrix} \alpha^* \\ \beta^* \end{bmatrix}$
- then $|\psi\rangle \langle\phi| = \begin{bmatrix} \alpha \\ \beta \end{bmatrix} [\alpha^* \beta^*] = \text{projector}$
- projector 'P' = $|\psi\rangle \langle\psi| \mapsto$ projector onto a state $|\psi\rangle$
- projectors $\rightarrow$ required in measurement theory.
- projectors $\mapsto$ probability of collapsing a quantum state into a specific basis state observed

Matrix representation of operators-few examples

- the following are same / as good as Gates only (pauli matrices)
- identity operator $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$
- pauli-X operator (NOT) gate operator $\sigma_x = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$
- Pauli-Z operator $\sigma_z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$

Functions of vectors and matrices



scalar valued functions



quadratic forms



includes VQE, regression functions

what and why of Functions of vectors and matrices

- in addition to reals, complex numbers, functions can include those with domains and co domains consisting of vectors and matrices—called 'functional analysis'— widely used in [quantum computing, quantum algorithms etc](#)
- among other 5 popular "scalar valued functions"— quadratic forms are of interest in which VQE etc arise.
- quadratic form $\xrightarrow{\text{yields}}$ a scalar.
- for a Hermitian Matrix 'A', the quadratic form $\langle x | A | x \rangle$ is always real, irrespective of $|x\rangle$ converse is also true. $\implies$ if $\langle x | A | x \rangle$ is real for any $|x\rangle$, A is necessarily Hermitian

- in quantum phenomenon, statistical average of measuring an observable represented by Hermitian matrix A on a quantum state $|\psi\rangle = \langle\psi| A |\psi\rangle$, and by mathematical theorem no necessary here— average values is real always i.e quadratic form must be real

- if A is Hermitian, then $g(x)$ is real where $g(x) = \langle x | A | x \rangle$, A being a square matrix.
- VQE relates to finding minimum of $g(x)$ under the constraint $\langle x | x \rangle = 1$
- alternately $\tilde{g}(x = \left(\frac{\langle x | A | x \rangle}{\langle x | x \rangle} \right))$ is minimized, only if A is Hermitian.

VQE in terms of 'H' Hamiltonian, as hybrid algorithm

- VQE is a hybrid quantum-classical algorithm to find smallest eigen value (ground state energy) of a quantum system
- more useful for Material science and quantum chemistry domains.
- essence of VQE is to minimize the expectation value of the Hamiltonian of the system on lines of quadratic form.
- formulation of the problem is $E(\theta) = \langle \psi(\theta) | H | \psi(\theta) \rangle$
- where H–Hamiltonian matrix of the system, $|\psi(\theta)\rangle \mapsto$ quantum state parametrized by θ ; quantum state vector is always normalized.

- VQE uses / depends on an iterative process—quantum computer $\xrightarrow{\text{evaluates}}$ $E(\theta)$ + classical computer optimizes parameter θ to find minimum $E(\theta)$
- such process benefits from strengths of both quantum and classical computing paradigms.
- there by efficiently solving problems that are otherwise challenging for classical systems alone.
- $E(\theta)_{min} \rightarrow$ ground state energy (corresponding eigenvalue of the $\mapsto$ ground state)
- in simplest case, VQE has $\rightarrow$ analytical solution,
- VQE (variational eigen solver) is used to solve multidimensional θ with much more complicated $|x\rangle$ and X

Regression coefficients as an example of scalar valued (vector) functions

- regression function is a vector function of the form
$$h(x) = \|Ax - y\|^2, |y\rangle \in \mathcal{C}^m, |x\rangle \in \mathcal{C}^n, A \in \mathcal{C}^{m \times n}$$
- in linear regression $A \mapsto$ matrix of predictors (design matrix), $|x\rangle \mapsto$ vector of coefficients to be estimated, $|y\rangle \mapsto$ vector of observations. $h(x) \mapsto$ squared norm of the residual vector $|Ax - y\rangle$.
- to find $|Ax - y\rangle_{min}$ is the purpose of linear regression.
- is equivalent to least squares fit problem

single qubit ket notation summary in computational basis

$$|\psi\rangle = \begin{bmatrix} z_0 \\ z_1 \end{bmatrix} = z_0 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + z_1 \begin{bmatrix} 0 \\ 1 \end{bmatrix} = z_0 |0\rangle + z_1 |1\rangle$$

z_0, z_1 — complex numbers obeying

$$|z_0|^2 + |z_1|^2 = 1$$

quantum state(multi qubit) = mathematical sum
(ket notation)

$$\psi_k = \sum_{k=0}^{2^n-1} Z_k |k\rangle$$

=

$$\sum_{k=0}^{2^n-1} (a_k + ib_k) |k\rangle$$

=

$$\sum_{k=0}^{2^n-1} \sqrt{p_k} (\cos \theta_k + i \sin \theta_k) |k\rangle$$

'n'— number of qubits; Amplitude

- i.e for 'k'th outcome, Amplitude $= a_k + ib_k$
- 'n qubit ' state outcomes $0 \leq k < 2^n$
- direction $= \theta_k$, probability p_k

quantum gates summary / essential details

Gate	Description
Pauli X 	It rotates the qubit 180 degrees in the X-axis. Maps $ 0\rangle$ to $ 1\rangle$ and $ 1\rangle$ to $ 0\rangle$. Also known as the bit flip or NOT gate. It is represented by the matrix: $X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$
Pauli Y 	It rotates around the Y-axis of the Bloch sphere by π radians. It is represented by the Pauli matrix: $Y = \begin{bmatrix} 0 & -j \\ j & 0 \end{bmatrix}$ where $j = \sqrt{-1}$ is known as the imaginary unit.
Pauli Z 	It rotates around the Z-axis of the Bloch sphere by π radians. It is represented by the Pauli matrix: $Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$
Hadamard 	It represents a rotation of π on the axis $(X + Z) / \sqrt{2}$. In other words, it maps the states: $0\rangle$ to $(0\rangle + 1\rangle) / \sqrt{2}$ $1\rangle$ to $(0\rangle - 1\rangle) / \sqrt{2}$
Phase $\sqrt{Z}$ 	This gate is required to make superpositions. It has the property that it maps $X \rightarrow Y$ and $Z \rightarrow Z$. This gate extends H to make complex superpositions.
Transposed Conjugate of S 	It maps $X \rightarrow -Y$ and $Z \rightarrow Z$.

Figure: from a reputed publication

Gate	Description
Controlled NOT (CNOT)	This is a two qubit gate that flips the target qubit (applies Pauli X) if the control is in state 1. This gate is required to generate entanglement
	
Phase $\sqrt{S}$	The $\sqrt{S}$ gate performs a halfway rotation of a two-qubit swap. It is universal such that any gate can be constructed from only $\sqrt{\text{swap}}$ and single qubit gates. It is represented by the matrix:
	$\sqrt{S} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1/2(1+i) & 1/2(1-i) & 0 \\ 0 & 1/2(1-i) & 1/2(1+i) & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$
Transposed Conjugate of T or Dagger-T	Represented by the matrix:
	$\sqrt{S}^\dagger = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1/2(1-i) & 1/2(1+i) & 0 \\ 0 & 1/2(1+i) & 1/2(1-i) & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$
Barrier	It prevents transformations across its source line
	
Measurement	The measurement gate takes a qubit in a superposition of states as input and spits either a 0 or 1. There is a probability of a 0 or 1 as output depending on the original state of the qubit. Always remember that measurement should be the last thing done in the circuit
	
Conditional	Conditionally apply a quantum operation
	
Identity	The identity gate performs an idle operation on the qubit for a time equal to one unit of time
	

Figure: from a reputed publication

Name	Description
Bra-ket vector notation	$\langle 0 = [1 \ 0] \quad 1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$
Bra = Row vector $\langle a $	
Ket = column vector $ a\rangle$	$ 0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad 1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$
Bra is the adjoint (complex conjugate transpose) of a ket	
Basis vectors and standard basis are linearly independent, they cannot be written as a linear combination of other vectors	Two qubit basis vectors:
Any vector can be written as a linear combination of basis vectors: $ \psi\rangle = c_1 00\rangle + c_2 01\rangle + c_3 10\rangle + c_4 11\rangle$	$ 00\rangle = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, 10\rangle = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}, 01\rangle = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}, 11\rangle = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$
Inner product $\langle a b\rangle$ results in a scalar	$\langle 0 0\rangle = [1 \ 0] \begin{bmatrix} 1 \\ 0 \end{bmatrix} = 1$
Outer product $ a\rangle\langle b $: Can be thought as an operator that transforms $ b\rangle$ to $ a\rangle$	$ 0\rangle\langle 0 = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, 1\rangle\langle 1 = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$ $ 1\rangle\langle 0 = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, 0\rangle\langle 1 = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$
Unitary matrix: It is a square complex matrix whose adjoint equals its inverse	$U = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, U^\dagger = \begin{bmatrix} a^* & c^* \\ b^* & d^* \end{bmatrix}, U U^\dagger = U^\dagger U = I$
	Hermitian: $U = U^\dagger$
State vector (wave function) $ \psi\rangle$	$ \psi\rangle = \sum_i \alpha_i e_i\rangle, \sum_i \alpha_i ^2 = 1$
Superposition	$ \psi\rangle = \alpha 0\rangle + \beta 1\rangle$ where $ \alpha ^2 + \beta ^2 = 1$

Figure: from a published book

Name	Description
Tensor product of two operators $A \otimes B$	
$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, B = \begin{bmatrix} e & f \\ g & h \end{bmatrix}$ $A \otimes B = \begin{bmatrix} a \begin{bmatrix} e & f \\ g & h \end{bmatrix} & b \begin{bmatrix} e & f \\ g & h \end{bmatrix} \\ c \begin{bmatrix} e & f \\ g & h \end{bmatrix} & d \begin{bmatrix} e & f \\ g & h \end{bmatrix} \end{bmatrix}$	$A \otimes B = \begin{bmatrix} ae & af & be & bf \\ ag & ah & bg & bh \\ ce & cf & de & df \\ cg & ch & dg & dh \end{bmatrix}$
Entangled vs separable states: Entangled states cannot be expressed as the tensor product of independent qubit states	$ \Psi\rangle = \frac{1}{\sqrt{2}} 00\rangle + \frac{1}{\sqrt{2}} 11\rangle (\text{entangled})$ $ \Psi\rangle = \frac{1}{\sqrt{2}} 01\rangle + \frac{1}{\sqrt{2}} 11\rangle (\text{separable})$
Eigenstate and eigenvalue	$M a\rangle = \lambda a\rangle$ λ (lambda) is the eigenvalue for eigenstate
Pauli operators	$X 0\rangle = 1\rangle, X 1\rangle = 0\rangle$ $Y 0\rangle = i 1\rangle, Y 1\rangle = -i 0\rangle$ $Z 0\rangle = 0\rangle, Z 1\rangle = - 1\rangle$
Measurement probability for state $ a\rangle$	$ \langle a \psi\rangle ^2$
Note: $\langle a \psi\rangle$ is the probability amplitude	
Expected value $\langle\psi\rangle$ after repeated measurements is always different	$\langle\psi\rangle = \sum \lambda_i \text{Pr}(a_i\rangle) = \langle\psi A \psi\rangle$
Unitary transformation	$ \psi_2\rangle = U \psi_1\rangle$



Applying H to the basis states $|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $|1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$:

$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

$$H|0\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \frac{1}{\sqrt{2}} \left(\begin{bmatrix} 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right) = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

$$H|1\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \frac{1}{\sqrt{2}} \left(\begin{bmatrix} 1 \\ 0 \end{bmatrix} - \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right) = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

Figure: from a published book

qubit rep'n / Bloch sphere

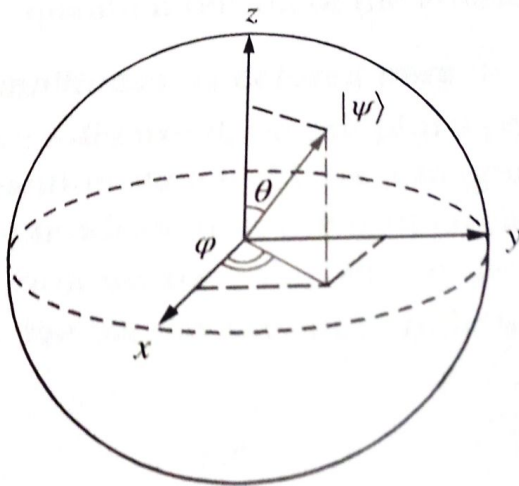


Figure: from a publication

Bloch sphere-slightly different

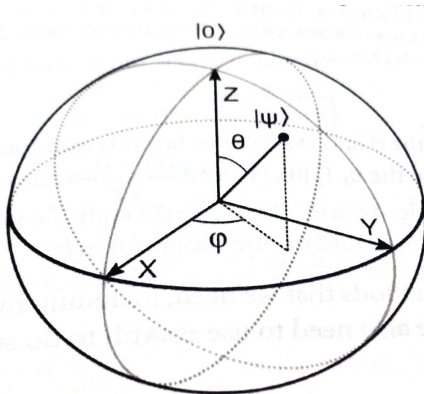


Figure: from a printed source with thanks to author